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Commentary

Do Short Delays in Treatment Affect Uveal Melanoma Prognosis?

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The timing of cancer treatment weighs heavily on the minds of patients and physicians. For patients with uveal melanoma (UM), treatment preparations, baseline systemic staging, and logistical arrangements often require several weeks or longer, especially for patients treated with plaque brachytherapy or proton beam radiotherapy. Although there is little doubt that long delays of a year or more likely increase the risk of metastatic death from UM, short intervals of a few weeks or months are not thought to meaningfully affect prognosis, because most UMs are estimated to have been present for years before diagnosis.[1]

In this study, Stålhammar[2] (p. 1094) performed a retrospective analysis of 1145 patients with UM between 2012 and 2022 to compare the incidence of UM-specific mortality in patients with prompt treatment (<1 month from diagnosis) versus delayed treatment (>1 month). Using multivariate competing risks analysis, the risk for UM-specific mortality was estimated to increase by 1% for every additional 10-day delay in treatment. These findings are proposed to "challenge the prevailing theory of early metastatic seeding in UM" and to signify "late metastatic seeding."

The author is to be commended for undertaking this large study to raise awareness of an important question. Its claims have far-reaching consequences that could lead to increasing patient anxiety, enucleations (which can be done faster than radiotherapy), treatment of benign lesions, lawsuits, and health care costs. Consequently, this study warrants careful appraisal.

Despite attempts to demonstrate that the prompt and delayed treatment groups are similar, the retrospective, uncontrolled study design makes it difficult to control for all prognostically significant variables, such as overall health and socioeconomic status. Indeed, the much higher non−UM-associated mortality in the delayed treatment group in Figure 6A (in the original article) suggests that the groups are *not* balanced with respect to significant survival variables. Of further concern, there was not an adequate stratification for genetic biomarkers, which are by far the most important determinants of metastatic risk in UM.[3]

The decision to use 4 weeks as the cutoff between prompt and delayed treatment is arbitrary and results in a skewed distribution in which the early group is relatively homogenous with respect to time to treatment, whereas the delayed group is highly heterogenous, ranging from 5 weeks to 3 years. Further, the choice of 10 days as the interval for assessing regression trend was arbitrary. Of even greater concern, the regression trend was assumed to be linear across time, but this is most likely a faulty assumption. The reported hazard of 1% per 10-day delay could in fact be infinitesimally small for short delays and much greater for long delays. The author could have assessed such a nonlinear relationship by modeling the effect of delay by a smoothing spline.

The use of UM-specific mortality is potentially problematic, because it can be difficult to define and ascertain this end point. This is why the Collaborative Ocular Melanoma Study and most other such studies have used overall survival as the primary end point.[4] Indeed, the apparent difference in mortality between prompt and delayed treatment groups in Figure 6C (in the original article) is greatly reduced and likely nonsignificant if considering overall mortality, which can be determined by summing the "UM mortality" and "other mortality" lines.

To support the claim of late metastatic seeding, great emphasis is placed on a "pronounced surge" in UM mortality after year 10 in patients with stage I UM in the delayed treatment group (Fig 6A in the original article). However, this claim is based on a mere 12 patients. Furthermore, these events occurred near the end of the follow-up period, where there are fewer at-risk individuals and more censored data points that can lead to biased survival estimates.

Uveal melanoma metastasis is not a one-time event but an ongoing, inefficient process with tumor cells being detectable in the circulation early in tumorigenesis, even in patients who do not develop metastasis.[5] Metastatic propensity is determined largely by a few key genetic aberrations, which usually occur early in tumor evolution in a punctuated burst.[6,7] The genetic evidence does not indicate that "metastases that leave the eye in later stages of the disease are at a higher risk of establishing growing macrometastases." Although long treatment delays likely increase the risk of metastasis (and thus why we treat UM), the inefficiency of the metastatic process and the slow doubling time[1] make it highly improbable that a delay of 10 days, or even several months, would have a clinically meaningful impact on mortality risk.

In conclusion, Stålhammar[2] presents an important and thought-provoking study that should stimulate further investigation. However, these findings alone should not be used as a rationale for choosing enucleation over eye-sparing therapy or for treating indeterminate lesions. Until such time that compelling evidence is provided to the contrary, it remains reasonable to presume that delays of up to several months in treating UM, and initial observation of indeterminate lesions, are unlikely to have a significant impact on survival.

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